



Asset Management System

Sponsor Proposal — Year 1 Implementation & Annual Operations

Prepared for the PNG Immigration & Citizenship Authority

AT A GLANCE

REFERENCE	NPAMS-ICSA-2026-001
ISSUED	11 May 2026
VALIDITY	60 days from issue
YEAR 1 TOTAL	PGK 104,500.00 incl. GST
VS REFERENCE	PGK 38,053.13 saving (~ 26.7%) vs Zyntrix v1.2
TIME TO LIVE	8 weeks from contract signature
USERS	15 concurrent web users included
HOSTING	PNG-resident SaaS, autoscaling, 99.5% SLA

Prepared by

NPAMS Programme Office · Independent State of Papua New Guinea

COMMERCIAL IN CONFIDENCE

1. Executive Summary

The PNG Immigration & Citizenship Authority (ICSA) requires a single, auditable system of record for the high-value assets it operates on behalf of the State — blank ePassport stocks, visa stickers, biometric workstations, secure printers, inspection kits and the supporting estate at headquarters and provincial border posts. The National Public Asset Management System (NPAMS) has been built specifically for the PNG public sector and is already live, multi-tenant, and integrated with the Government's geographic data set.

This proposal sets out the activities, timeline and commercials required to onboard ICSA onto NPAMS as an Authority tenant alongside other PNG agencies already provisioned (PNGDF, RPNGC, IRC, Treasury, Customs, Ombudsman). It delivers the same outcomes as the previously-tabled Zyntrix proposal — a modern, role-based, GIS-enabled asset register with purchase, audit and maintenance workflows — at materially lower total cost.

Total programme cost (Year 1, incl. GST)	PGK 104,500.00
Comparable Zyntrix proposal (incl. GST)	PGK 142,553.13
Savings to ICSA in Year 1	PGK 38,053.13 (~ 26.7%)
Time to go-live	8 weeks from signature
Concurrent web users included	15

NPAMS is delivered as a hosted, autoscaling PNG-resident SaaS platform with a defined Service Level Agreement, encrypted data at rest, RBAC scoped down to facility level, and tamper-evident HMAC-SHA256 digital signatures on every purchase, approval and goods-receipt event.

2. About NPAMS

NPAMS is the State-sponsored asset management platform for PNG. It is currently operating in production for the following Authorities, each in its own logical tenant with its own branding, scope and roles:

- PNG Defence Force (PNGDF)
- Royal Papua New Guinea Constabulary (RPNGC)
- Internal Revenue Commission (IRC)
- Department of Treasury
- PNG Customs Service
- Office of the Ombudsman
- PNG Immigration & Citizenship Authority (this proposal)

Because NPAMS is multi-tenant from the database upward, ICSA does not pay for a new build — only for its onboarding, configuration, migration, training, hosting and ongoing support. There is no platform IP licence fee charged to ICSA above and beyond the per-user subscription described in §6.

3. Asset Categories in Scope

NPAMS organises every ICSA asset into one of three top-level groups, aligned to the Zyntrix taxonomy. Each group is configurable and ICSA Asset Officers can add sub-classes without vendor intervention.

Group	Sub-classes	Examples
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Group	Sub-classes	Examples
Digital & IT	ICT Hardware, Office Equipment, Software Licences, Network & Cyber assets	Biometric capture stations, fingerprint & document scanners, secure printers, servers, workstations, laptops, photocopiers, switches / routers / firewalls, software entitlements
Physical & Operational	Secure Stock, Furniture & Fittings, Border Post Estate, Uniforms & PPE	Blank ePassport booklets (32 / 64-page), visa stickers, security inks, holographic foils, office furniture, secure cabinets and safes, buildings, perimeter fencing, signage, inspection lanes, officer uniforms, body armour, inspection PPE
Mobile & Distributed	Vehicles & Plant, Mobile Biometric Kits, Field Equipment, Generators / UPS	Pool vehicles, patrol vehicles, marine craft, generators, UPS units, ruggedised mobile enrolment kits, handheld document readers deployed across border posts and provincial offices

Every asset record carries: unique asset code, category, location (province -> district -> facility), custodian, acquisition cost, depreciation, condition, service history, photographs, and a public verification QR code.

4. Solution Scope

The following modules are already implemented in NPAMS and will be enabled for ICSA on day one. Live screenshots are included in Annex A.

4.1 Multi-Agency Tenancy & RBAC

- Logical isolation per Authority (ICSA tenant pre-provisioned).
- Implemented role catalogue: Super Admin, National Asset Controller, National Auditor, Provincial Admin, Provincial Asset Officer, Provincial Viewer, and Agency Admin (per-Authority).
- Scope is enforced at every API call: national, agency, province, district or single facility. ICSA users see only ICSA data.

4.2 Asset Register

- Full lifecycle: acquisition -> in-service -> maintenance -> disposal.
- Bulk import via spreadsheet template; per-asset photographs and documents.
- Public QR verification page (no login) for field officers and auditors.

4.3 Stock & Inventory (secure consumables)

- On-hand quantity, reorder level, supplier and unit cost per stock item.
- Stock movements (receive / issue / transfer / adjust) with full audit trail.
- Per-facility balances; automated low-stock alerts.

4.4 Purchase Request Workflow with Digital Signatures

- Officer raises a request -> routed to scoped approvers -> approve / reject -> goods receipt against the request.
- Every state transition is signed. The signer types their full name; the server stores signed_name, signed_at and a tamper-evident signed_hash = HMAC-SHA256(server_secret, "v1[user|action|request|timestamp|name]").
- Notifications dispatched to requester and approvers at every step.

4.5 Audit Sessions

- Scheduled or ad-hoc audit cycles by location.

- Officers verify each asset (present / missing / damaged) and capture evidence photographs against the public QR code.
- Variance report generated on session close.

4.6 Maintenance & Service History

- Preventive and corrective maintenance scheduling per asset.
- Cost capture, downtime tracking, vendor records.

4.7 GIS Map (PNG-specific)

- Interactive Leaflet map of all 22 provinces and districts (GADM boundaries).
- Province / district selection dims unrelated areas; OSM, satellite and topographic basemaps.
- Used to visualise asset and stock distribution across ICOSA's national footprint and border posts.

4.8 Reports & Dashboard

- KPI dashboard per role; custody, condition, depreciation and movement reports.
- CSV / PDF export.

4.9 Notifications & Activity Log

- In-app notification centre delivering events to requesters and approvers.
- Append-only activity_logs table capturing user, action type, entity, description and timestamp for every CRUD and workflow event.
- Workflow events on signed actions additionally carry the actor's HMAC-SHA256 signature, providing tamper-evident proof independent of the database row.

5. Governance, Security & Compliance

Area	NPAMS Control
Authentication	Bcrypt-hashed passwords; JWT session tokens with short TTL; lockout after repeated failures.
Authorisation	Role-Based Access Control with scope filtering at the data layer.
Digital signatures	HMAC-SHA256 keyed by server secret on every purchase / approval / receipt event — verifiable, non-repudiable, replay-resistant.
Audit log	Append-only activity_logs and purchase_request_events tables; workflow events additionally signed by the actor with HMAC-SHA256.
Data residency	Hosted in PNG-accessible cloud region; daily encrypted backups retained 30 days.
Data in transit	TLS 1.2+ end-to-end.
Data at rest	AES-256 encryption at the storage layer.
PII	Limited to officer profile data needed for workflow attribution; no citizen biometric data is processed.
Disaster Recovery	RPO 24h, RTO 8h. Documented runbook handed over at Go-Live.
Service Level	99.5% monthly uptime; P1 1h, P2 4h, P3 next business day.

Area	NPAMS Control
Vulnerability management	Monthly dependency CVE scan (npm audit + Snyk); SAST run on every release; critical CVEs patched within 7 days, high within 30 days; quarterly external penetration test summary delivered to ICSA.

NPAMS aligns with the PNG Government Digital Strategy and is designed to be auditable by the Auditor-General's office on demand.

6. Commercial

All amounts in Papua New Guinea Kina (PGK). GST applied at 10% per the Goods and Services Tax Act 2003.

6.1 Year 1 Cost Build-Up

#	Line item	Unit	Unit Price	Qty	Line Total (PGK)
1	Platform Licence — Year 1	Named concurrent web user / year	600.00	15	9,000.00
2	Implementation & Configuration	Fixed-price work package (one-off)	28,000.00	1	28,000.00
3	Data Migration & Seeding	Fixed-price work package (one-off)	8,500.00	1	8,500.00
4	User Training	Classroom / online session	2,500.00	3	7,500.00
5	Cloud Hosting (12 months)	Hosting bundle / month (app + DB + CDN + backups)	1,000.00	12	12,000.00
6	Annual Support & Maintenance — Year 1	SLA support bundle / month (incl. <= 8 hrs/mo enhancements)	2,500.00	12	30,000.00
Subtotal (excl. GST)					95,000.00
GST (10%)					9,500.00
Grand Total Year 1 (inci. GST)					104,500.00

6.2 Recurring Cost — Year 2 onwards

#	Line item	Unit	Unit Price	Qty	Line Total (PGK)
1	Platform Licence	Named concurrent web user / year	600.00	15	9,000.00
2	Cloud Hosting	Hosting bundle / month	1,000.00	12	12,000.00
3	Annual Support & Maintenance	SLA support bundle / month	2,500.00	12	30,000.00
Subtotal (excl. GST)					51,000.00
GST (10%)					5,100.00
Recurring Total / yr (inci. GST)					56,100.00

Additional concurrent users may be added at any time at PGK 600 / user / year (prorated). No volume

change requires a re-contract.

6.3 Three-Year Total Cost of Ownership

Year	Incl. GST (PGK)
Year 1	104,500.00
Year 2	56,100.00
Year 3	56,100.00
3-Year TCO	216,700.00

7. Implementation Timeline

Eight (8) weeks from contract signature to formal Go-Live, with two further weeks of post-go-live hyper-care.

Workstream	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Mobilisation										
Tenant Provisioning										
Data Migration										
Configuration										
Integration										
Training										
UAT										
Go-Live										
Hyper-care										

8. Roles & Responsibilities

Activity	NPAMS	ICSA
Tenant provisioning	R	I
Data extraction from legacy systems	C	R
Data cleansing & sign-off	C	R
Configuration of roles, scopes, categories	R	C
User Acceptance Testing	C	R
Training delivery	R	C
Go-Live decision	C	R
Day-2 support	R	C

R = Responsible · C = Consulted · I = Informed

9. Assumptions & Exclusions

- Pricing assumes 15 concurrent web users in Year 1; additional users billable at PGK 600 / user / year.
- Mobile applications (Android / iOS) are not in scope; the responsive web app works on tablets and modern phones.
- Hardware (scanners, label printers, biometric kiosks) is provided by ICSA.
- Connectivity to provincial border posts is provided by ICSA.
- Custom development beyond the 8 hrs/month support envelope is quoted separately at PGK 300 / hr.

10. Terms & Conditions

- 1. Currency. All amounts are in PGK and exclude GST unless otherwise stated.
- 2. Payment. 30% on contract signature, 40% on UAT sign-off, 30% on Go-Live. Recurring fees billed annually in advance.
- 3. Validity. This proposal is valid for sixty (60) days from the date on the cover.
- 4. Variations. Any change in scope is captured in a written Change Request and priced at PGK 300 / hr.
- 5. Intellectual Property. Platform IP remains with the NPAMS Programme. ICSA data remains the property of ICSA at all times and is exportable on demand in open formats.
- 6. Termination. Either party may terminate for convenience on 90 days' notice. On termination, all ICSA data is delivered as Postgres dump and CSV export within 14 days at no charge.
- 7. Confidentiality. Both parties treat the contents of this engagement as confidential.
- 8. Governing Law. Laws of the Independent State of Papua New Guinea.

Annex A — Live Application Screenshots

The following screenshots are taken from the running production build of NPAMS on the date of issue and reflect the system that ICSA will inherit.

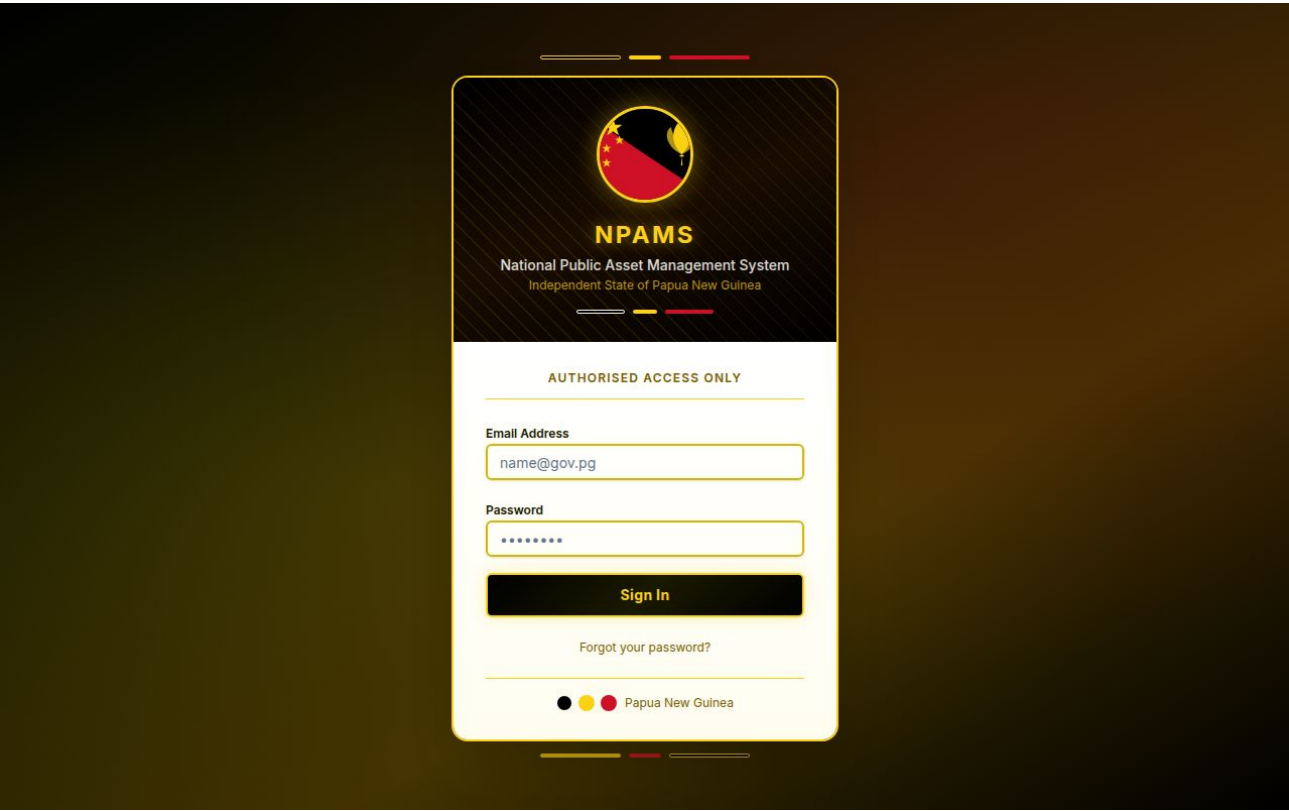


Figure A1 — Sign-in screen with PNG branding and tenant theming.

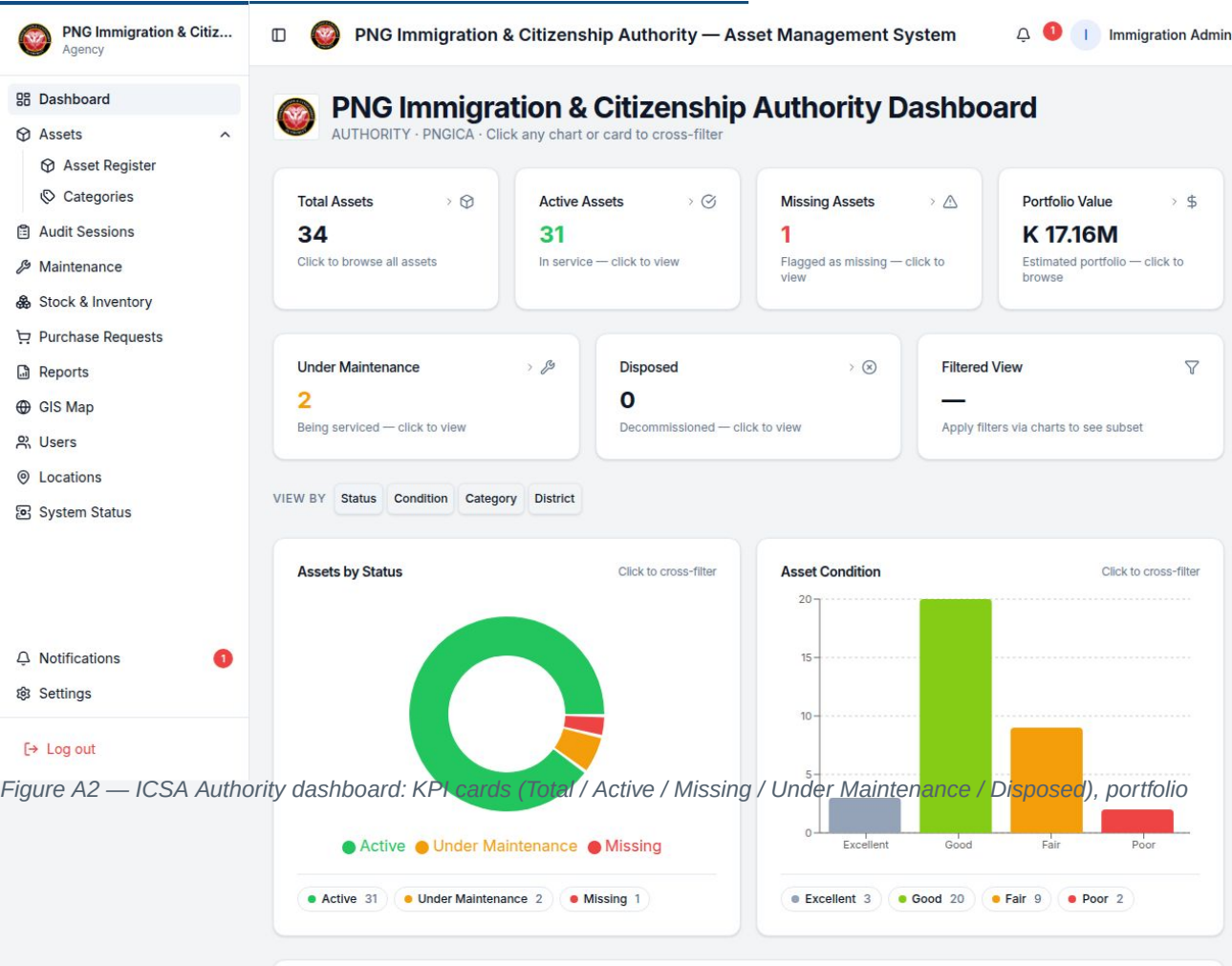


Figure A2 — ICSA Authority dashboard: KPI cards (Total / Active / Missing / Under Maintenance / Disposed), portfolio

value, status & condition charts and the most-recently added assets, all scoped to the PNGICA tenant.

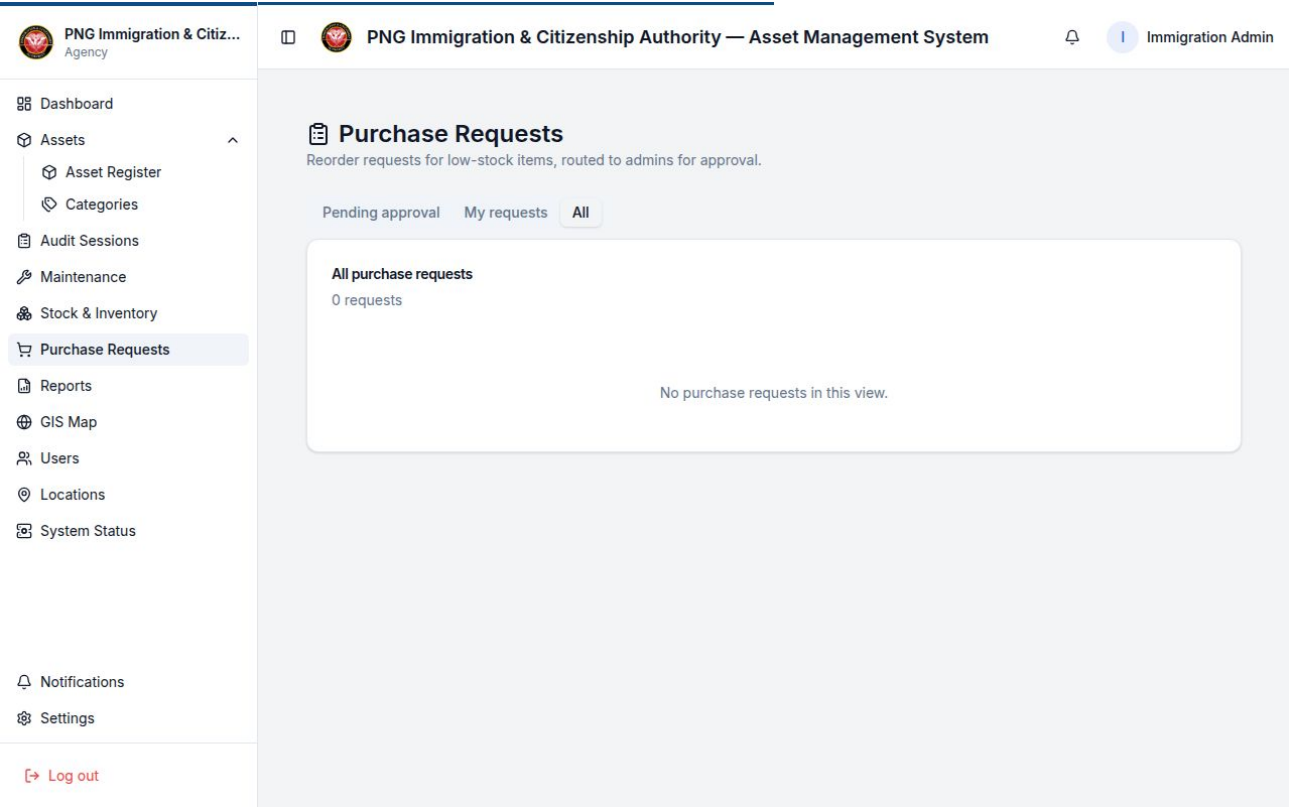


Figure A3 — Purchase Requests workflow page — the digital signature pipeline that ICSCA will use to govern stock replenishment of ePassport booklets, visa stickers and other secure consumables.

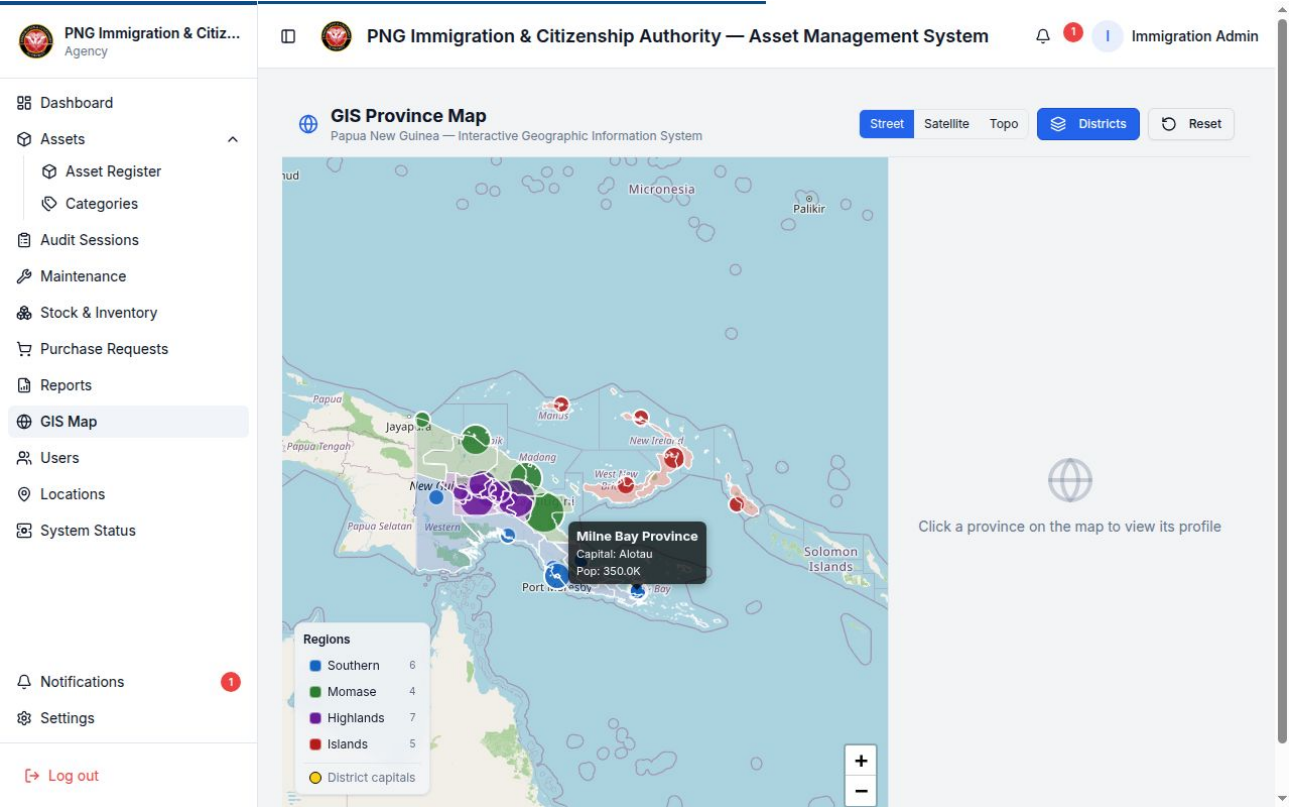


Figure A4 — GIS Province Map: interactive Leaflet map of all 22 PNG provinces with district overlays, basemap selector and per-province asset / population profile (Street / Satellite / Topo).

Annex A1 — Signature ledger evidence (live data)

The two events below were generated by the running NPAMS API on 11 May 2026 against purchase request PR-20260511-5092 (Blank ePassport Booklet, qty 500 @ K 42.50 from Crane Currency PNG, requester Immigration Admin, role Agency Admin). The signed_hash for each event is the actual HMAC-SHA256 value persisted in the purchase_request_events table. ICSA's Auditor can re-compute these hashes from the documented input string and the server secret to prove that no event has been forged or tampered with after the fact.

SUBMITTED

signed_at 2026-05-11 19:26:45 UTC

signer: Immigration Admin · role: Agency Admin

signed_hash: **de2ff7ef0f18e1212980ed820079bf7f869ab7620d91dbf5382bb7ad6da60e8a**

APPROVED

signed_at 2026-05-11 19:26:45 UTC

signer: Immigration Admin · role: Agency Admin

signed_hash: **afaaa07ffbba697d56087029da25564066eaf0b0c62b536ba9d7bac596da1c4da**

Annex B — Comparison vs Reference Proposal

Item	Zyntrix v1.2	NPAMS	Diff (PGK)
Platform licence (Year 1, 5 vs 15 users)	42,000.00	9,000.00	(33,000.00)
Implementation & customisation	included	28,000.00	—
Data migration & seeding	included	8,500.00	—
Annual hosting	included	12,000.00	—
Training	included	7,500.00	—
Annual support & maintenance	84,000.00	30,000.00	(54,000.00)
Per-user licence (additional)	718.75	600.00	(118.75)
Subtotal (excl. GST)	129,593.75	95,000.00	(34,593.75)
GST (10%)	12,959.38	9,500.00	(3,459.38)
Grand Total Year 1 (incl. GST)	142,553.13	104,500.00	(38,053.13)
Saving to ICSA	—	—	~ 26.7%

The NPAMS proposal is itemised line-by-line so ICSA's Finance team can trace every Kina; nothing is bundled into an opaque "support" line.

End of proposal — Commercial in Confidence — NPAMS Programme Office, 2026.